

RADU ANDREI TUDORICA

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EDUCATION

National University of Science and Technology POLITEHNICA Bucharest Sept 2024 – Present
Bachelor of Science in Computer Science (ACS CTI) Bucharest, Romania

- **Relevant Coursework:** Algorithm Analysis, Data Structures, Computer Programming (C), Operating Systems, Assembly Language & Computer Architecture, Computer Networks, Communication Protocols, Numerical Methods

EXPERIENCE

Google Summer of Code 2026 – Open-Source Contributor May 2026 – Sept 2026
Unikraft Remote

- Ported and updated 6 application library ports (Redis, Nginx, SQLite, Lua, Python3, libgo) to the current Unikraft unikernel core, resolving build failures, patch conflicts, and missing syscalls.
- Identified and fixed concrete build blockers for Redis 8: dropped a redundant patch, rewrote a second patch for a changed function signature, added 12 new source files and a C++ dependency to the build system.
- Updated Nginx from 1.15 to 1.28, corrected moved HTTP/2 source paths in `Makefile.uk`, and confirmed successful boot inside QEMU.
- Contributed 3 merged PRs to Unikraft prior to GSoC, covering unit tests and error handling for `ukalloc`, `ukallocbuddy`, and `ukallocpool`.

PROJECTS

ACSQL – In-Memory Database & Query Engine | *C, Memory Management, Cryptography* | [GitHub](#) May 2024 – Jun 2024

- Developed an in-memory relational database management system in C, supporting complex entity relationships and dynamic schema management.
- Implemented a custom query engine with support for filtered searches, conditional updates, and cascading deletes across related tables.
- Engineered a **Cipher Block Chaining (CBC)** encryption module from scratch using bitwise operations and custom S/P-box substitution tables.
- Achieved zero memory leaks across all operations – verified via **Valgrind** with rigorous tracking of nested dynamically allocated structures.

Minecraft Chunk Engine (libchunk) | *C, Advanced Pointers, Bitwise Optimization* | [GitHub](#) Nov 2024 – Dec 2024

- Built a C library for 3D spatial data manipulation, representing Minecraft chunks via multi-dimensional pointer structures (`char***`).
- Implemented a custom **Run-Length Encoding (RLE)** compression algorithm using advanced bitwise shifts and masking to reduce memory footprint.
- Engineered recursive matrix transformations for gravity simulation and procedural generation of 3D geometries (spheres, cuboids).
- Managed complex dynamic memory reallocation routines with zero leaks across all execution paths.

Lox Interpreter – Java Implementation | *Java, Compiler Architecture, AST, Parsing* | [GitHub](#) Jul 2025 – Present

- Built a complete interpreter from scratch: lexical scanner, recursive descent parser, and tree-walk evaluator.
- Implemented runtime support for variable scoping, first-class functions, closures, and control flow.
- Applied the **Visitor Pattern** for clean separation between AST structure and evaluation logic.

TECHNICAL SKILLS

Languages: C, C++, Bash, Java, Scala, Python, x86 Assembly, MATLAB
Build & Debug Tools: GCC, Makefile, Valgrind, GDB, Maven, QEMU
Developer Tools: Git, GitHub, VS Code, Linux, Remote Access Tools
Concepts: Memory Management, Systems Programming, Unikernel Architecture, Compiler Design, Data Structures & Algorithms
Scripting & Automation: Bash scripting, CLI tool development